

EE5630 Digital Signal Processing

Lecturer: Yi-Wen Liu

Homework #5: Butterworth Filter Design Flow

Due 17:00 Wednesday Nov. 24 2021

1. Write a function that selects the **appropriate** T_d and returns DEN and NUM automatically for any given order N and cut-off frequency Ω_c in $(0, \pi)$.
2. For instance, construct a 6^{th} -order filter with $\Omega_c = 0.2\pi$.
3. In the **same plot, compare** the magnitude responses for $N = 4 \sim 20$.
4. What is the minimum N if we require the magnitude response at 0.28π to be smaller than 0.01?
5. Continuing from question 2, modify your function so that it selects the **appropriate** order N automatically given the edge frequency (e.g., 0.28π) of the stop band and the stop band tolerance (e.g., 0.01). You can assume that the cut-off frequency for the pass band is 0.2π .
6. Write a function that automatically turns the resulting [NUM, DEN] from question 5 into a difference equation so that the output $y[n]$ can be computed iteratively given a input $x[n]$.
7. Use the difference equation to filter an audio file. Again, you may assume that the cut-off frequency for the passband is 0.2π .

This homework requires **precise answers** and **build from scratch** (means write everything in your own mathematical way).

Forbidden built-in matlab functions - filter(), conv(), butter(), buttord(), functions which are related to create a filter directly

Butterworth filter

$$|H_c(j\Omega)|^2 = \frac{1}{1 + \left(\frac{\Omega}{\Omega_c}\right)^{2N}}, \quad \begin{array}{ll} 1 \leq |H(e^{j\omega})| \leq 1, & 0 \leq |\omega| \leq 0.2\pi \\ |H(e^{j\omega})| < 0.01, & 0.28\pi \leq |\omega| \leq \pi \end{array}$$